

MP1994V2 Milestone Tracker

Mortensen–Pissarides (1994) Lean replication

31 July 2026

Last completed milestone: M6 – Unemployment flow balance and the full static steady state – **COMPLETE – AMBER**

Next milestone: M7 – Global comparative statics – NOT STARTED

Source of truth. docs/milestone_tracker.md. Status vocabulary: NOT STARTED; IN PROGRESS; BLOCKED; READY FOR REVIEW; COMPLETE – GREEN; COMPLETE – AMBER; DEFERRED / OUTSIDE CORE LEAN. M0 through M4 are complete and green. M5 is complete and amber: uniqueness is green and existence is conditional on an additional bracket. M5b is unstarted. M6 is complete and amber overall: its stock completion and uniqueness are green, while existence inherits M5’s amber qualification. M7 has not started.

Milestone plan

The tracker uses one readable block per milestone instead of a dense landscape table. Each block contains the same fields as the Markdown table.

M0. Foundations and primitive value equilibrium

Status: **COMPLETE – GREEN**

Paper: Sections 2–3; equations (1)–(7)

Economic target: Primitives, layered assumptions, probability/matching definitions, and primitive value equilibrium.

Expected declarations: Primitives; four assumption bundles; `cdf`; meeting rates; `surplus`; `continuationTerm`; `ValueEquilibrium`

Adequacy grade: GREEN **Dependencies:** None

Blocker: None

Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed M0 interface while beginning M1 separately.

M1. Surplus Bellman equation

Status: **COMPLETE – GREEN**

Paper: Section 3, journal p. 401; equation (8)

Economic target: Derive equation (8) from equations (3)–(7).

Expected declarations: Continuation decomposition; `firm_share`; `surplus_flow_equation`; `surplus_bellman_of_probability`; `surplus_bellman`

Adequacy grade: GREEN **Dependencies:** M0 complete

Blocker: None

Branch/commit or tag; last review; next action: `issue2`; completion commit; 2026-07-29 human mathematical/economic review; preserve the reviewed conditional representation theorem.

M2. Affine surplus and endogenous reservation cutoff**Status:** COMPLETE – GREEN**Paper:** Section 3; reservation discussion and equation (12)**Economic target:** Surplus difference theorem, strict monotonicity, unique endogenous reservation threshold, and equation (12).**Expected declarations:** Surplus difference, affinity, strict monotonicity, zero uniqueness, reservation theorem**Adequacy grade:** GREEN **Dependencies:** M1**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-29 human mathematical/economic review; preserve M2 interfaces and design the measure-form continuation identity for M3.**M3. Continuation value, job destruction, and job creation****Status:** COMPLETE – GREEN**Paper:** Section 3; equations (9), (10), and (13)**Economic target:** Measure-form continuation identity and derivation of job destruction and job creation.**Expected declarations:** expectedExcess; tailOptionValue; equations (9), (10), and (13); JD/JC residual predicates; M3 capstone**Adequacy grade:** GREEN **Dependencies:** M2**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-30 human mathematical/economic review; preserve the independent JD/JC interfaces and neutral capstone module.**M4. Reduced-equilibrium reconstruction and equivalence****Status:** COMPLETE – GREEN**Paper:** Section 3; equations (10), (13); journal p. 402**Economic target:** Two-way representation between primitive values and the admissible reduced pair.**Expected declarations:** ReducedEquilibrium; generic tail identity; forward and reverse maps; round trips; nonempty equivalence**Adequacy grade:** GREEN **Dependencies:** M3**Blocker:** None**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-31 human mathematical/economic review; preserve representation interfaces and formulate corrected continuity, endpoint, and crossing assumptions for M5.**M5. Existence and uniqueness of static equilibrium****Status:** COMPLETE – AMBER**Paper:** Section 3; equations (10), (13), Figure 1, journal p. 402**Economic target:** Existence and uniqueness of the static reduced equilibrium.**Expected declarations:** Expected-excess regularity; JD/JC slopes; StaticExistenceAssumptions; conditional unique existence; value-side economic uniqueness**Adequacy grade:** Uniqueness: GREEN; existence: AMBER; overall: AMBER **Dependencies:** M4**Blocker:** Existence uses a lower positive crossing condition not derived from primitives**Branch/commit or tag; last review; next action:** issue2; completion commit; 2026-07-31 human review; preserve the conditional theorem and see the M5b design in docs/static_existence_foundation.md. This does not block M6.

M5b. Primitive foundation for static existence**Status:** NOT STARTED**Paper:** Foundational refinement to Section 3 static existence**Economic target:** Replace the assumed lower crossing with a derived result.**Expected declarations:** Proposed matching-boundary bundle; upper-job profitability; derived lower_crossing; optional Cobb–Douglas certification**Adequacy grade:** Target: existence GREEN under a strengthened general theorem or certified Cobb–Douglas specialization **Dependencies:** M5**Blocker:** The Inada condition on q alone is insufficient; JD-implied tightness must become positive through a separate profitability condition**Branch/commit or tag; last review; next action:** No branch/tag or review; add right-hand Inada and upper-job profitability, derive StaticExistenceAssumptions, and see docs/static_existence_foundation.md. M5b is not M6.**M6. Unemployment flow balance and full static steady state****Status:** COMPLETE – AMBER**Paper:** Section 3; equation (14), Figure 2, journal pp. 403–404**Economic target:** Derive unemployment, employment, vacancies, and the full static state.**Expected declarations:** Strict mass/CDF bridge; hazards and flows; steadyStateUnemployment; SteadyStateEquilibrium; equation (14); exact reduced/full equivalence**Adequacy grade:** Stock completion/equation (14): GREEN; uniqueness: GREEN; existence: AMBER inherited from M5; overall: AMBER **Dependencies:** M4–M5**Blocker:** No new M6 blocker. Full-state nonemptiness inherits M5's unproved lower-crossing foundation**Branch/commit or tag; last review; next action:** issue2; M6 completion commit; 2026-07-31 human mathematical/economic review; preserve the split capstones and adequacy contract; M7 is NOT STARTED.**M7. Global comparative statics****Status:** NOT STARTED**Paper:** Section 3**Economic target:** Order-theoretic comparative statics for p, b, λ, r, σ where possible without differentiability.**Expected declarations:** Parameter-order theorems; equilibrium order lemmas**Adequacy grade:** – **Dependencies:** M5**Blocker:** Global solution selection and parameter monotonicity**Branch/commit or tag; last review; next action:** No branch/tag or review; separate global monotone results from derivatives.**M8. Appendix derivative comparative statics****Status:** NOT STARTED**Paper:** Appendix; equations (A1)–(A12)**Economic target:** Differentiable equilibrium paths and derivative sign results.**Expected declarations:** Analytic assumption layer; Jacobian/implicit-path theorems**Adequacy grade:** – **Dependencies:** M5, M7**Blocker:** Differentiability and nondegenerate Jacobian**Branch/commit or tag; last review; next action:** No branch/tag or review; keep Appendix assumptions out of the core.**M9. Two-state anticipated cyclical shocks****Status:** NOT STARTED**Paper:** Section 4; equations (15)–(31)**Economic target:** Two-state surplus systems, cutoff ordering under explicit hypotheses, and impact asymmetry.**Expected declarations:** Two-state value system; ordering theorems; employment-measure impact results**Adequacy grade:** – **Dependencies:** M3–M6**Blocker:** Piecewise continuation and employment measure**Branch/commit or tag; last review; next action:** No branch/tag or review; begin with an explicit two-state process.

M10. Finite Markov equilibrium and employment accounting**Status:** NOT STARTED**Paper:** Analytical prelude to Section 5; equations (32)–(38)**Economic target:** Finite-state equilibrium representation, employment transition operator, creation/destruction accounting, and equation (38).**Expected declarations:** Finite kernel; transition operator; mass preservation; accounting identity**Adequacy grade:** – **Dependencies:** M6, M9**Blocker:** Finite-state interface and numerical-solver connection**Branch/commit or tag; last review; next action:** No branch/tag or review; prefer finite state before general measurable states.**NUM. Numerical simulation and calibration****Status:** DEFERRED / OUTSIDE CORE LEAN**Paper:** Section 5; equations (39)–(42), Tables I–II**Economic target:** External Python or Julia numerical replication; Lean only for exact accounting or certified residuals where useful.**Expected declarations:** External solver and optional Lean residual/accounting certificates**Adequacy grade:** – **Dependencies:** Analytical interfaces as needed**Blocker:** Numerical/empirical work is outside core Lean**Branch/commit or tag; last review; next action:** No branch/tag or review; define a reproducible calibration and residual protocol externally.**Adequacy notes.** M6: docs/static_steady_state_adequacy.md. Inherited M5 existence foundation: docs/static_existence_foundation.md.**M0 acceptance checklist**

- ☒ New module root is isolated from the legacy architecture.
- ☒ Primitive parameter structure exists.
- ☒ Economic assumptions are layered.
- ☒ Shock law is represented by a measure.
- ☒ CDF is derived from the measure.
- ☒ Matching rates are defined.
- ☒ Surplus is defined from J, W, U , not stored independently.
- ☒ Equations (1)–(7) are represented faithfully.
- ☒ Equations (5) and (6) use one shared continuation term.
- ☒ No cutoff or reduced-equilibrium conclusion is assumed.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No `sorry`, `admit`, or custom axioms exist.
- ☒ Tracker TeX compiles and PDF has been visually inspected.
- ☒ Human mathematical/economic review is complete.

M1 acceptance checklist

- ☒ Generic continuation decomposition is proved under a probability measure.
- ☒ Firm surplus share is derived from equations (3) and (4).
- ☒ The surplus flow equation is derived from equations (5)–(7).
- ☒ Equation (8) is proved for the actual equilibrium surplus.
- ☒ The main theorem uses only probability normalization and existing integrability.
- ☒ The **ShockAssumptions** wrapper installs the probability instance locally.
- ☒ No M0 structure was strengthened or modified.
- ☒ No cutoff, affine surplus, or reduced equilibrium was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, or custom axioms exist.
- ☒ Proof ledger TeX/PDF compiles and passes visual inspection.
- ☒ Milestone tracker TeX/PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M2 acceptance checklist

- ☒ Equation (8) is subtracted at two states.
- ☒ The scaled two-point surplus identity is proved.
- ☒ The divided affine-surplus identity is proved.
- ☒ Surplus is proved strictly increasing.
- ☒ The cutoff is defined from actual equilibrium surplus.
- ☒ The cutoff is proved to be a surplus zero.
- ☒ The zero is proved unique.
- ☒ $V = 0$ is derived from free entry and $r > 0$.
- ☒ $q(\theta)J(\varepsilon_u) = c$ follows from (1)–(2).
- ☒ $J(\varepsilon_u) > 0$ is proved.
- ☒ $S(\varepsilon_u) > 0$ is proved.
- ☒ $\varepsilon_d < \varepsilon_u$ is proved.
- ☒ Equation (12) is proved.
- ☒ Surplus signs are characterized relative to ε_d .
- ☒ Firm-value signs are characterized relative to ε_d .
- ☒ Active surplus is rewritten as a scaled positive part.
- ☒ No M0 or M1 mathematical declaration was changed.
- ☒ No M3 or later theorem was introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, or custom axioms exist.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M3 acceptance checklist

- ☒ Expected excess is the integral of the positive part.
- ☒ Tail option value is defined from the induced CDF.
- ☒ Expected-excess integrability is proved.
- ☒ Strict upper-tail probability equals one minus the CDF.
- ☒ The strict-tail layer-cake identity is proved.
- ☒ Active-surplus expectation is rewritten using expected excess.
- ☒ Measure-form equation (9) is derived.
- ☒ Paper-facing equation (9) is derived.
- ☒ The free-entry search-gain identity is derived.
- ☒ Measure-form equation (10) is derived.
- ☒ Paper-facing equation (10) is derived.
- ☒ Job-destruction residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-destruction predicate.
- ☒ The product-form job-creation identity is derived.
- ☒ Paper equation (13) is derived.
- ☒ Job-creation residual and predicate are defined.
- ☒ The equilibrium pair satisfies the job-creation predicate.
- ☒ No **ReducedEquilibrium** structure is introduced.
- ☒ No M0–M2 mathematical declaration is changed.
- ☒ No M4 or later theorem is introduced.
- ☒ Targeted builds pass.
- ☒ Full build passes.
- ☒ No **sorry**, **admit**, custom axiom, or opaque placeholder exists.
- ☒ Proof-ledger PDF compiles and passes visual inspection.
- ☒ Tracker PDF compiles and passes visual inspection.
- ☒ Human mathematical/economic review is complete.

M4 acceptance checklist

- ☒ Robust product-form job-creation predicate is defined.
- ☒ Reduced pair stores positive theta and an admissible cutoff.
- ☒ Measure-form JD and product-form JC are the only stored equations.
- ☒ Generic positive-part integrability and tail identity are proved.
- ☒ Reduced paper equations (10) and (13) are derived.
- ☒ Every value equilibrium maps to a reduced equilibrium.
- ☒ Candidate surplus, J, U, W, V, and explicit Nash wage are defined.
- ☒ Active surplus is integrable and candidate equation (8) is proved.
- ☒ Reconstructed equations (1)–(7) are proved.
- ☒ A reduced pair reconstructs a value equilibrium and its cutoff.
- ☒ Reconstruct–reduce is exact; reverse economic components agree.
- ☒ Conditional nonempty equivalence is proved.
- ☒ No unconditional existence, uniqueness, M0–M3 change, or M5 theorem.
- ☒ Targeted and full builds pass; audits and PDFs pass.
- ☒ Human mathematical/economic review is complete.

M5 acceptance checklist

- ☒ Expected excess is antitone, one-Lipschitz, and continuous.
- ☒ JD net value and JD-implied tightness are defined.
- ☒ JD satisfaction is equivalent to $\theta = \theta_{JD}(d)$.
- ☒ The JD curve is strictly increasing.
- ☒ The scalar JC residual is defined and strictly decreasing on its admissible domain.
- ☒ Figure 1 JD-upward and JC-downward slope theorems are proved.
- ☒ At most one `ReducedEquilibrium` is proved.
- ☒ `StaticExistenceAssumptions` contains no root or equilibrium witness.
- ☒ Bracket continuity and upper residual $-c$ are proved.
- ☒ An interior root and a reduced witness are constructed by the IVT.
- ☒ Conditional unique reduced existence is proved.
- ☒ Value nonemptiness, theta/cutoff uniqueness, and economic uniqueness follow through M4.
- ☒ Existence is labeled conditional on the additional bracket assumptions.
- ☒ No M0–M4 theorem statement changed and no M6 theorem was introduced.
- ☒ Targeted/full builds, audits, and both PDFs pass.
- ☒ Human mathematical/economic review is complete.

M6 acceptance checklist

- ☒ Strict destruction mass matches the M2 rule and equals the CDF under no atoms.
- ☒ Separation/job-finding hazards, creation/destruction flows, and the residual are defined.
- ☒ Total transition hazard is positive.
- ☒ Closed-form unemployment is in $[0, 1]$, balances flows, and is unique.
- ☒ Paper equation (14) is proved.
- ☒ `SteadyStateEquilibrium` is defined; employment and vacancies are derived.
- ☒ Creation is $q(\theta)v$; destruction is $\lambda F(d)(1 - u)$; flows are equal.
- ☒ Positive separation implies positive unemployment and vacancies.
- ☒ $v/u = \theta$ is proved only under positivity.
- ☒ Every reduced equilibrium completes to a full state; exact round trips and an equivalence are proved.
- ☒ At most one full state is proved without existence assumptions.
- ☒ Conditional nonemptiness and unique existence are transported from M5.
- ☒ No Beveridge-curve slope, M0–M5 statement change, or M7+ theorem was introduced.
- ☒ Targeted/full builds, no-sorry/axiom audits, and prohibited-pattern searches pass.
- ☒ Proof-ledger and tracker PDFs compile and pass visual inspection.
- ☒ Human mathematical/economic review is complete.

M5b foundational follow-up

Status: NOT STARTED. The adequacy target is to upgrade static existence from amber to green under either a strengthened general matching-boundary theorem or a fully certified Cobb–Douglas specialization. M5b is a foundational refinement, not M6, and is not part of the current M5 Lean proof.

The proposed work adds a right-hand Inada condition for q , separately requires $b < p + \sigma \varepsilon_u$, proves that the JD curve reaches zero and then positive tightness, derives `lower_crossing`, and constructs `StaticExistenceAssumptions`. The Inada condition on q alone is insufficient because it says nothing about whether JD-implied tightness ever becomes positive. The complete design is in `docs/static_existence_foundation.md`. M6 does not implement this refinement.

Changelog

2026-07-31 – M6 human review completed. Stock completion, equation (14), accounting, exact representation, and full-state uniqueness were judged mathematically and economically faithful and are GREEN. Zero separation correctly yields $u = v = 0$, with the literal ratio guarded by positive separation. M6 adds no existence closure premise; existence inherits M5's `lower_crossing` and remains AMBER. Added split green/amber capstones and the detailed record `docs/static_steady_state_adequacy.md`. M6 is **COMPLETE – AMBER**; M7 is **NOT STARTED**.

2026-07-31 – M6 unemployment flow balance implemented. Added the strict destruction mass and atomless CDF bridge, hazards, flows, the unique balanced unemployment formula, and equation (14). Added the full static state with derived employment/vacancies, guarded $v/u = \theta$ by positive separation, and proved exact reduced/full equivalence, uniqueness, and conditional existence inherited from M5. Stock completion is recommended GREEN; existence and overall M6 were submitted as AMBER pending the completed review above.

2026-07-31 – M5 static-existence foundation documented. Recorded the exact current continuity and lower-crossing assumptions and the unimplemented M5b route combining right-hand Inada behavior with separate upper-job profitability. The note makes explicit that Inada alone is insufficient. M5 remains **COMPLETE – AMBER**; M6 remains **NOT STARTED**. The durable design note is `docs/static_existence_foundation.md`.

2026-07-31 – M5 review completed. The uniqueness and IVT proofs passed review. Figure 1 establishes uniqueness, not existence. The non-circular `lower_crossing` field assumes a positive residual rather than a root, but supplies the missing endpoint condition. Existence-only signatures now omit matching assumptions. Uniqueness is **COMPLETE – GREEN**; existence and overall M5 are **COMPLETE – AMBER**. A future task is to derive the lower crossing from primitive conditions on q , or certify it for Cobb–Douglas; this does not block M6.

2026-07-31 – M5 static existence and uniqueness implemented. Expected excess is one-Lipschitz; JD slopes upward and JC downward, proving at most one reduced equilibrium. The separate continuity/lower-bracket bundle and the derived upper residual $-c < 0$ yield an interior crossing by the IVT. M4 transports conditional existence and economic uniqueness to value equilibria. The implementation was submitted for review; equation (14) remains M6.

2026-07-31 – M4 review completed. Review found the generic excess and CDF-tail analysis valid, and equations (10) and (13) sufficient for reconstruction under the stated admissibility assumptions. The Nash wage is economically correct; equations (1)–(7) are reconstructed; and both economic round trips are certified. The import diagram and assumption-field account were corrected. M4 proves no existence or uniqueness theorem and is **COMPLETE – GREEN**.

2026-07-30 – M4 reconstruction and equivalence implemented. Created the robust reduced pair, generic analytic layer, forward bridge, explicit reverse reconstruction, and equivalence module. Equations (1)–(7) are reconstructed with the Nash wage; the reduced round trip is exact and the value round trip is componentwise exact. Nonemptiness equivalence consumes a witness on either side and proves no existence or uniqueness. M4 is **READY FOR REVIEW**.

2026-07-30 – M3 review completed. The layer-cake/CDF proof was reviewed and found valid without atomlessness. Equations (9), (10), and (13) have the correct signs and dependencies, and job creation is independent of the job-destruction equation. The capstone was moved to the neutral combining module `SteadyState/StaticConditions.lean`. Residual predicates require separate admissibility fields in M4, which must also prove a generic expected-excess/tail theorem for reverse reconstruction. M3 was graded **COMPLETE – GREEN**.

2026-07-30 – M3 continuation and static conditions implemented. Created `SteadyState/Continuation.lean`, `SteadyState/JobDestruction.lean`, and `SteadyState/JobCreation.lean`, with the capstone in the neutral `SteadyState/StaticConditions.lean`. Expected excess follows from the M2 active-surplus formula; Mathlib's strict-tail layer-cake theorem gives the CDF representation without atomlessness. Equations (9), (10), and (13), the two residual predicates, and the forward M3 capstone are proved for an existing `ValueEquilibrium`. No reduced-equilibrium structure or M4 theorem was introduced.

2026-07-29 – M2 review completed. Review confirmed that equation (8) was correctly subtracted at two states and that actual equilibrium surplus is affine and strictly increasing. The derived `reservationCutoff` is its uniquely identified zero; equations (1)–(2) establish $\varepsilon_d < \varepsilon_u$; and equation (12) and the reservation rule are proved. M2 remains conditional on an existing `ValueEquilibrium` and was graded **COMPLETE – GREEN**.

2026-07-29 – M2 affine surplus and cutoff implemented. Created `SteadyState/AffineSurplus.lean` and `SteadyState/Cutoff.lean`. Subtracting equation (8) gives affine surplus and strict monotonicity. The derived cutoff is the unique surplus zero; equations (1)–(2), positive vacancy cost and meeting, and $\beta < 1$ separately place it below ε_u . Added equation (12), reservation sign rules, the active-surplus bridge, and the M2 capstone. M2 is **READY FOR REVIEW**.

Review-archive convention. Beginning with M3, project-local review archives are named `review_m3.zip`, `review_m4.zip`, and so on. They remain untracked and are never committed.

2026-07-29 – M1 review completed. Human mathematical and economic review found equation (8) mathematically faithful to the paper and its proof non-circular. It is derived from equations (3)–(7), uses the actual equilibrium surplus, and assumes no cutoff, affinity result, or reduced equilibrium. It is a conditional representation theorem for an existing `ValueEquilibrium`; M1 does not prove equilibrium existence. M1 was graded **COMPLETE – GREEN**.

2026-07-29 – M1 equation (8) implemented. Created `MP1994V2/SteadyState/Surplus.lean` with the continuation decomposition, firm share, surplus flow equation, minimal probability theorem, and `ShockAssumptions` wrapper. Updated `All`, `Audit`, `README`, trackers, and added the proof ledger. M1 is **READY FOR REVIEW**.

2026-07-29 – M0 review completed. Review found equations (1)–(7) economically faithful and no future conclusion embedded; corrected substantive modules $\rightarrow$ `All` $\rightarrow$ `Audit`; required explicit `ShockAssumptions` `P` and local `D.isProbability` in M1; graded **COMPLETE – GREEN**.

2026-07-29 – v2 foundation created. Added `Definitions`, `Assumptions`, `Probability`, `Matching`, `Equilibrium/Value`, `All`, `Audit`, `README`, and synchronized trackers: primitives, four assumption bundles, CDF/support/matching notation, shared surplus/continuation, `ValueCandidate`, and `ValueEquilibrium`. No legacy v1 Lean or architecture file changed; the library-root import makes `lake build certify MP1994V2.All`.